ic_autolab round 24 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:55:52Z at git `d7fa7145f82849c0b91d4ff65e5162bc3382fbbc`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-66c1d2` (`python3 tools/pcc/pcc.py standing PCC-20261104-66c1d2`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**improvement**	wsum2-b30-n2	0.900 [0.845, 0.959]	0.909 [0.847, 0.975]	0.957 [0.861, 1.064]	sel < inc	3.318 / 3.573 / 5.863	3.687 / 3.931 / 6.128	1.222 / 1.332 / 2.326	2.7x / 2.7x / 2.5x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

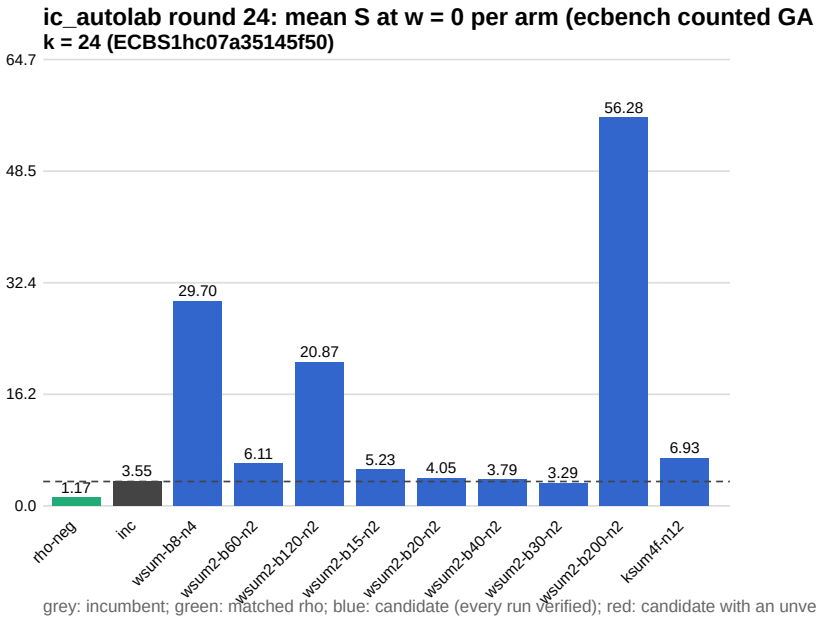
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.9001 [0.845, 0.959] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 24: session `ECBS1hc07a35145f50` (run exit 0, verify exit 0, receipt sha256 `8cf5ec24c85c69f22a2d1cb100bf344425d0ddec867fe4012ad0e1c7e8cd6af9`)

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.168 / 1.274 / 2.220	matched reference
`inc`	wsum-b8	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=8,free=1`	32	32	3.547 / 3.764 / 5.720	incumbent
`wsum-b8-n4`	wsum-b8	4	`prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=8,free=1`	32	32	29.701 / 29.725 / 29.941	incumbent neighbourhood: N = 1.2 x 3
`wsum2-b60-n2`	wsum2-b60	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	6.110 / 6.175 / 6.764	cross-size transfer: wsum2-b60-n2 is the k = 28 incumbent at N / N_a = 0.010
`wsum2-b120-n2`	wsum2-b120	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	20.867 / 20.885 / 21.044	cross-size transfer: wsum2-b120-n2 is the k = 32 incumbent at N / N_a = 0.004
`wsum2-b15-n2`	wsum2-b15	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	5.225 / 5.995 / 12.923	exploration: family wsum2-b15 untried at k = 24, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`wsum2-b20-n2`	wsum2-b20	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=20,free=1`	32	32	4.049 / 4.566 / 9.219	exploration: family wsum2-b20 untried at k = 24, N = 0.0001 x N_a
`wsum2-b40-n2`	wsum2-b40	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=40,free=1`	32	32	3.792 / 3.949 / 5.360	exploration: family wsum2-b40 untried at k = 24, N = 0.0001 x N_a
`wsum2-b30-n2`	wsum2-b30	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	3.292 / 3.542 / 5.793	exploration: family wsum2-b30 untried at k = 24, N = 0.0001 x N_a
`wsum2-b200-n2`	wsum2-b200	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=200,free=1`	32	32	56.284 / 56.292 / 56.367	exploration: family wsum2-b200 untried at k = 24, N = 0.0001 x N_a
`ksum4f-n12`	ksum4f	12	`prime-projected-abscissa:size=12,scan_cap=192`	`ksum:m=4,free=1`	32	32	6.926 / 7.442 / 12.085	exploration: family ksum4f untried at k = 24, at its parent ksum4's best N = 12



train screen

Selection

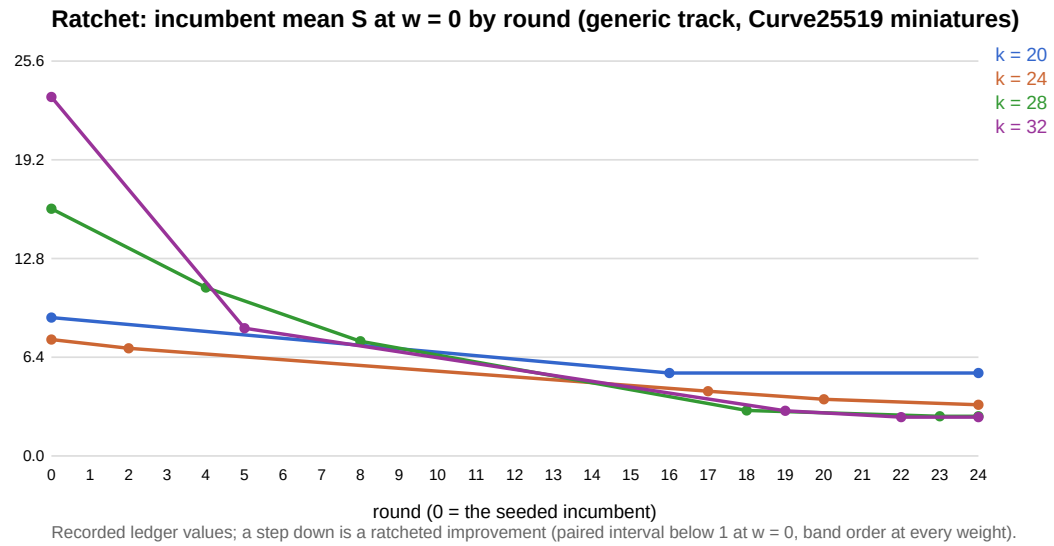
- k = 24: challenger `wsum2-b30-n2` (Pareto alternative at w = 1: `wsum2-b40-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
24	ECBS1h4a5add065b8e	0	0	24097ced646444e3eae4738baa175ec04c58eadd6fc8d5c798bdaf2bbc287c80	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	wsum-b5-n3 (prime-projected-abscissa:size=3,scan_cap=48 + wsum:coef=5,free=1)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	wsum2-b30-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=30,free=1)	24	3.318 / 3.573 / 5.863	2.7x / 2.7x / 2.5x
28	wsum2-b60-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=60,free=1)	23	2.576 / 2.793 / 4.746	2.8x / 2.7x / 2.6x
32	wsum2-b120-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=120,free=1)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.